

ReactCampus Work Completed Report

A full-stack college discovery & lead-management platform — comprehensive progress summary of all engineering work delivered across backend, frontend, and infrastructure.



As of June 4, 2026



Full-Stack MERN



Core Build Complete

~27,900 lines of code

20

Database models

140+

API endpoints

63

UI routes / pages

85

RBAC permissions

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Executive Summary

Where the project stands today.

ReactCampus is a production-grade college discovery platform comparable to CampusOption / Shiksha.com. The application is built as a monorepo with an **Express + MongoDB backend** and a **React 19 + TypeScript frontend**. The core platform is feature-complete: the public-facing portal, the full admin content-management suite, a dynamic form builder, a lead-management CRM, and a fine-grained role-based access system are all implemented and wired end-to-end.



Build status: core platform delivered

All 10 major feature areas are implemented across both tiers. The system is operational with seeded data, authentication, and the complete admin → public content pipeline working end to end.

Architecture Delivered

A clean, layered monorepo with a strict separation of concerns.



Backend — Express API

JavaScript · CommonJS · ~11,400 LOC across 137 files

- ✓ Layered architecture: Route → Controller → Service → Model
- ✓ 20 Mongoose models with indexes & geospatial support
- ✓ 21 route resources mounted under `/api/v1`
- ✓ Joi validation layer on every write endpoint
- ✓ Centralised error handling & standard response envelope
- ✓ Winston structured logging



Frontend — React SPA

TypeScript · Vite · ~16,400 LOC across 180 files

- ✓ Feature-based organisation — 26 feature modules
- ✓ Zustand auth store + TanStack Query data layer
- ✓ Axios client with JWT refresh-token rotation
- ✓ React Router 7 with permission-guarded routes
- ✓ Tailwind CSS design system, 19 reusable UI components
- ✓ React Hook Form + Zod validated forms

Backend Work Completed

Twenty data models, a complete REST API, and a hardened security layer.



Auth & Security

- ✓ JWT access + refresh tokens
- ✓ Refresh-token rotation with family-chain reuse detection
- ✓ bcrypt password hashing (12 rounds)
- ✓ Helmet, CORS, mongo-sanitize, HTML sanitization
- ✓ httpOnly secure cookies
- ✓ Append-only audit logging



RBAC Engine

- ✓ 85 granular permissions, 13 groups
- ✓ 4 system roles seeded
- ✓ AND / OR authorization middleware
- ✓ LRU permission cache
- ✓ Content-assignment scoped access



Catalog & Content

- ✓ Colleges, Courses, Exams, Categories, Cities
- ✓ Geospatial (2dsphere) + full-text search
- ✓ Polymorphic content sections
- ✓ College scoring / ranking / tiers
- ✓ CMS pages with dynamic college filters



Leads & CRM

- ✓ Full lead lifecycle & status history
- ✓ Assignment, notes, priority, tags
- ✓ Bulk actions, export, statistics
- ✓ Source tracking (form / channel)



Dynamic Forms

- ✓ Form builder, 10+ field types
- ✓ Conditional logic & validation rules
- ✓ Versioning + submission snapshots
- ✓ Lead field mapping, page assignment



Platform Services

- ✓ Reviews & discussions w/ moderation
- ✓ SEO metadata engine
- ✓ Dashboard & analytics aggregations
- ✓ Global site-settings config
- ✓ 14 database seed scripts

Frontend Work Completed

A complete admin console plus a public student-facing portal.



Admin Console — 18 modules, 52 routes

- ✓ Dashboard + dedicated Analytics view
- ✓ Users, Roles & permission management
- ✓ Colleges / Courses / Exams with content-section editors
- ✓ Categories, CMS Pages, SEO manager
- ✓ Form builder & submission viewer
- ✓ Lead list, lead detail & pipeline (Kanban) view
- ✓ Review & discussion moderation queues
- ✓ Content assignments, site settings, profile/security



Public Portal — 8 modules, 11 routes

- ✓ Homepage with hero, search & featured content
- ✓ College listing + rich college detail pages
- ✓ Course directory & course detail
- ✓ Exam listing & exam detail
- ✓ About, Contact, and Education-loan pages
- ✓ Dynamic CMS pages (/pages/:slug)
- ✓ Dynamic form renderer (/forms/:slug)
- ✓ Embedded reviews & discussion threads



UI Library

- ✓ 19 reusable components
- ✓ DataTable, Modal, Tabs, Toast
- ✓ Pagination, StatusBadge, StarRating



Content Editors

- ✓ Rich-text (Tiptap) editor
- ✓ Table, FAQ & List editors
- ✓ Discussion section component



Access Control

- ✓ Auth / Guest / Permission guards
- ✓ Per-route permission gating
- ✓ Mirror of backend permissions

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Feature Completion Matrix

Status of every major module across the full stack.

MODULE	SCOPE DELIVERED	PROGRESS	STATUS
Authentication & Sessions	JWT, refresh rotation, password change, profile		● Complete
RBAC & Permissions	85 permissions, 4 roles, guarded routes		● Complete
User & Role Management	Full CRUD, activation, role assignment		● Complete
College Catalog	CRUD, publish, geospatial, scoring, sections		● Complete
Course & Exam Catalog	CRUD, content sections, public detail pages		● Complete
Dynamic Form Builder	Builder, conditional logic, versioning, submissions		● Complete
Lead Management CRM	Lifecycle, pipeline, notes, export, stats		● Complete
CMS Pages & Content	Dynamic pages, filters, rich content blocks		● Complete
Reviews & Discussions	Submission, moderation queues, public display		● Complete
SEO Management	Per-entity meta, OG/Twitter, structured data		● Complete
Dashboard & Analytics	Stats, pipeline, activity feed, aggregations		● Complete
Public Portal	Home, listings, detail pages, contact, loan		● Complete
Site Settings & Config	Hero, featured, footer, contact, visibility		● Complete
Automated Testing	No formal test suite configured yet		● Pending
Media / File Uploads	Image URLs supported; native upload pipeline pending		● Partial

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By the Numbers

Measured scale of the delivered codebase.



Backend metrics

- ✓ 137 source files · ~11,400 lines of code
- ✓ 20 models · 19 controllers · 20 services
- ✓ 21 route resources · 140+ endpoints
- ✓ 50+ public read endpoints · 90+ protected
- ✓ 7 middleware layers · 14 seed scripts



Frontend metrics

- ✓ 180 source files · ~16,400 lines of code
- ✓ 26 feature modules
- ✓ 63 routes (52 admin + 11 public)
- ✓ 19 UI components · 4 content editors · 3 guards
- ✓ 20 typed entity definitions

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Technology Stack

Everything used to build the delivered system.



Frontend

React 19

TypeScript

Vite

Tailwind CSS

React Router 7

Zustand

TanStack Query

React Hook Form

Zod

Tiptap



Backend & Data

Node.js

Express

MongoDB

Mongoose

Joi

JWT

bcryptjs

Helmet

Winston

LRU-cache

Remaining & Next Steps

Items outside the delivered core, recommended for the next phase.



Test Coverage

Add an automated unit / integration test suite (none configured today) to lock in regression safety before scaling.



Media Pipeline

Native file/image upload & storage (S3 / CDN). Images are currently referenced by URL.



Deployment Hardening

Production env config, rate-limiting tuning, monitoring & CI/CD pipeline for release automation.